

	EMERGENCY MEDICAL SERVICES AGENCY	POLICY NUMBER	706
	A Division of the Merced County Department of Public Health	Effective Date	11/2025
Authority: Health and Safety Code, Division 2.5, and California Code of Regulations, Title 22, Division 9, Chapter 4, 1797.220		Initial Date:	07/2022
		Review Date:	10/2028

BRADYCARDIA	
ADULT	PEDIATRIC
BLS	
Assess Vitals / Obtain SpO ₂ / Oxygen - Titrate to SpO ₂ of 94% or higher / Secure Airway - Assist Ventilations if needed. If significant ALOC, accompanied with poor skin signs, initiate CPR (Pediatric Heart Rate less than 60 beats/minute).	
ALS	
Follow BLS procedures if applicable. Obtain 4-Lead ECG and 12-Lead ECG (12-Lead – Adult Only). Vascular Access – IV/IO rate as appropriate. Establish large bore IV if hypotensive – Refer to EMS Policy #738 – Non-Traumatic Shock. Capnography.	
Symptomatic Bradycardia	
Symptomatic Bradycardia is defined by ALL the following - Heart rate < 50 beats per minute - SBP < 90 mmHg - Associated signs and symptoms including ALOC, signs of shock, chest pain, acute pulmonary edema, syncope, extreme weakness	Symptomatic Bradycardia is defined by ALL the following - Heart rate < 60 beats per minute - SBP below normal limits per length-based assessment tape or application - Associated signs and symptoms including ALOC, signs of shock, chest pain, acute pulmonary edema, syncope, extreme weakness
Atropine - 0.5 mg IV/IO - Repeat every 3-5 minutes until SBP is > 90 mmHg - Max total dose 3 mg Atropine should not be used for patients in wide complex bradycardias or 2 nd or 3 rd degree heart blocks. If there is no improvement in the patient's condition,	Epinephrine 1:10,000 - 0.01 mg/kg IV/IO - Repeat every 3 minutes until hypotension is resolved per length-based assessment tape or application - If there is no response to Epinephrine, consider Atropine Atropine - 0.02 mg/kg IV/IO - Minimum dose 0.1 mg

Signatures on File

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consider transcutaneous pacing.

- Max dose 0.5 mg

Transcutaneous Pacing

Do not delay TCP if patient is severely symptomatic or difficulty establishing IV/IO access

Patients that need sedation or pain control to tolerate TCP, consider using **ONE** of the following Sedation:

Midazolam – 2 - 5 mg IV/IO

- Single dose

Pain Management – Morphine **OR** Fentanyl:

Morphine – 2 - 5 mg IV/IO

Fentanyl – 20 - 50 mcg IV/IO

May repeat once after 5 minutes if SBP > 90 mmHg.

Fentanyl or Morphine is preferred for patients with chest pain or suspected STEMI's.

Base Hospital Physician Orders Only

Transcutaneous Pacing (TCP) Pediatric

Patients that need sedation or pain control to tolerate TCP, consider using **ONE** of the following Sedation:

Midazolam - 0.1 mg/kg IV/IO

- Single dose

Pain Management – Morphine **OR** Fentanyl:

Morphine - 0.1 mg/kg IV/IO

Fentanyl - 1 mcg/kg IV/IO

May repeat once after 5 minutes if SBP is greater than length-based assessment tape or application target.

Special Considerations

If the patients' condition does **not** improve consider using Push Dose Epinephrine

Push Dose Epinephrine - 10mcg slow IV/IO (10mcg/ml)

- Repeat every 1-5 minutes until SBP is > 90mmHg